COMPSCI 572.01 (NLP)
Reasoning

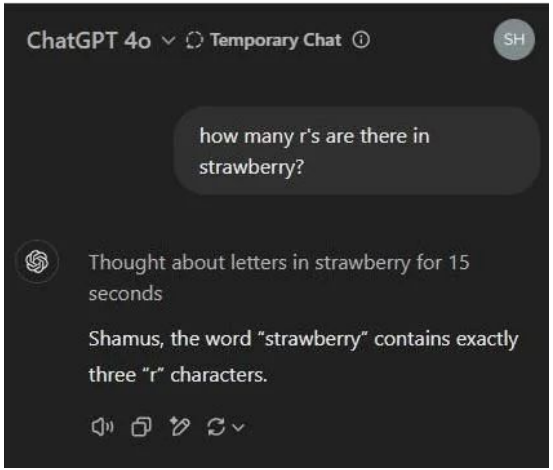
Reasoning models

Models which “think” before responding – by generating intermediate natural language sequences

This is most useful for difficult tasks which involve complex, multi-step reasoning

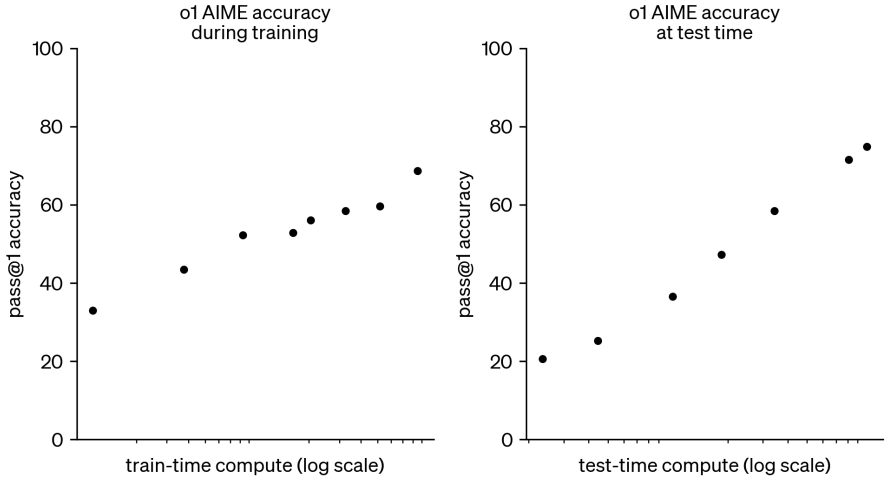
Expand the product $(x-2)^2(x+2)^2$. What is the product of the nonzero coefficients of the resulting expression, including the constant term?

This is a complex or challenging question, and it is difficult to provide a direct and correct answer. I need to think about it.
Well, so I need to expand $(x - 2)^2(x + 2)^2$ and then find the product of all the nonzero coefficients, including the constant term. Hmm, let’s see. Maybe there’s a smarter way to do this than just expanding both squares first and then multiplying them together? Wait, actually, I notice that $(x - 2)^2(x + 2)^2$ is the same as $[(x - 2)(x + 2)]^2$, right? Because when you have two squares multiplied, you can combine the bases first and then square the product. That might be easier because $(x - 2)(x + 2)$ is a difference of squares, which I remember is $x^2 - 4$. Let me check that: $(x - a)(x + a) = x^2 - a^2$, so yeah, a is 2 here, so that’s $x^2 - 4$. Then we need to square that result. Okay, so let’s write that down step by step to make sure I don’t make a mistake.
First step: $(x - 2)^2(x + 2)^2 = [(x - 2)(x + 2)]^2 = (x^2 - 4)^2$. Yep, that seems right. Now, let’s expand $(x^2 - 4)^2$. How do you square a binomial? $(a - b)^2 = a^2 - 2ab + b^2$, so here a is x^2 and b is 4. Let’s apply that formula. So, $(x^2)^2 - 2*(x^2)*(4) + (4)^2$. Let’s compute each term: $(x^2)^2$ is x^4, $2*x^2*4$ is $8x^2$, so with the negative sign it’s $-8x^2$, and 4^2 is 16. Wait, so putting that all together, is it $x^4 - 8x^2 + 16$? Let me verify by expanding the original expression the long way to make sure I didn’t skip a step wrong. Sometimes when combining terms early, you might miss something, but let’s check.
Original: $(x - 2)^2(x + 2)^2$. Let’s first compute $(x - 2)^2$ and $(x + 2)^2$ separately. $(x - 2)^2 = x^2 - 4x + 4$, right? Because $(a - b)^2 = a^2 - 2ab + b^2$, so $x^2 - 4x + 4$. Similarly, $(x + 2)^2 = x^2 + 4x + 4$, since $(a + b)^2 = a^2 + 2ab + b^2$. Now, we need to multiply these two quadratics together: $(x^2 - 4x + 4)(x^2 + 4x + 4)$. Maybe we can group terms here to make multiplication easier. Let’s see, notice that $(x^2 + 4)$ is a common part in both, and then one has $-4x$ and the other has $+4x$. So actually, this is $[(x^2 + 4) - 4x][(x^2 + 4) + 4x]$, which is again a difference of squares! Oh, that’s another way to see it, which is exactly how we got to $(x^2 - 4)^2$ earlier, because $(A - B)(A + B) = A^2 - B^2$ where $A = x^2 + 4$ and $B = 4x$? Wait, no, wait: hold on, if $A = x^2 + 4$ and $B = 4x$, then $A^2 - B^2 = (x^2 + 4)^2 - (4x)^2 = x^4 + 8x^2 + 16 - 16x^2 = x^4 - 8x^2 + 16$. Wait, that’s the same result as before! So whether we combine the original two squares into a single square of a product (which became $x^2 - 4$ squared) or combine the expanded squares into another difference of squares, we get the same quartic polynomial. Good, so that confirms that the expansion is indeed $x^4 - 8x^2 + 16$.
Let me just do the full multiplication of $(x^2 - 4x + 4)(x^2 + 4x + 4)$ the standard way to be 100% sure. Let’s distribute each term in the first polynomial across the second one.
First, multiply x^2 by each term in the second polynomial: $x^2*x^2 = x^4$, $x^2*4x = 4x^3$, $x^2*4 = 4x^2$



Test-time scaling

We can continue to improve model accuracies by generating longer and longer sequences at test time



OpenAI o1 (Sep 2024)

Outline

- Reasoning via prompting
  - Chain-of-thought
  - Self-consistency
- Reasoning via training
  - SFT
  - RLVR
  - PPO
  - GRPO

Chain-of-thought prompting

In-context learning with explanations added in the demonstrations

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Standard prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

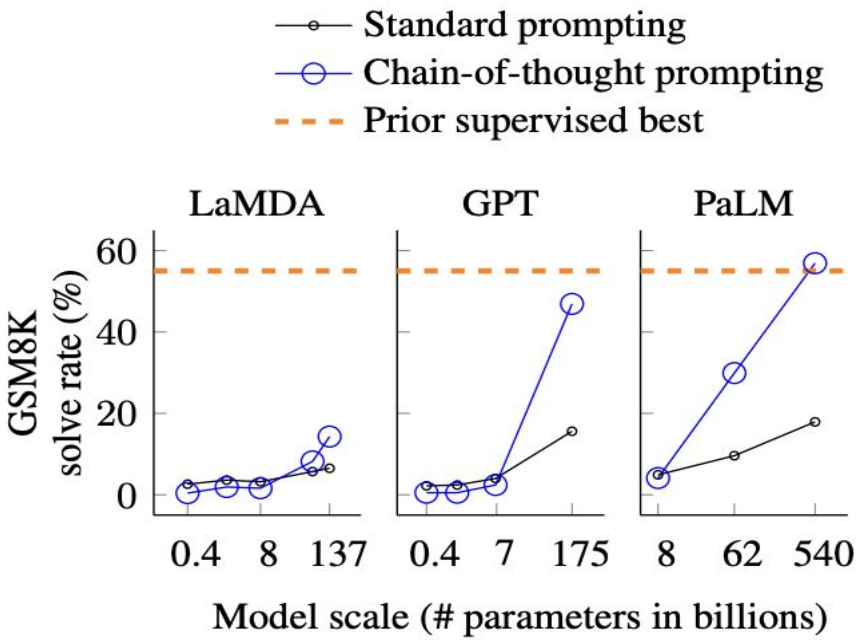
Chain-of-thought prompting

Wei et al; 2022

Chain-of-thought prompting

Chain-of-thought is also an emergent property of model scale

Middle-school math problems



Wei et al; 2022

Chain-of-thought prompting

More tasks where chain-of-thought helps

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

Last Letter Concatenation

Q: Take the last letters of the words in “Lady Gaga” and concatenate them.

A: The last letter of “Lady” is “y”. The last letter of “Gaga” is “a”. Concatenating them is “ya”. So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

These are also known as “reasoning traces”

Wei et al; 2022

Zero-shot chain-of-thought

We can also prompt the model to do chain-of-thought reasoning **without** providing labeled examples

Few-shot chain-of-thought

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

Zero-shot chain-of-thought

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Kojima et al; 2023

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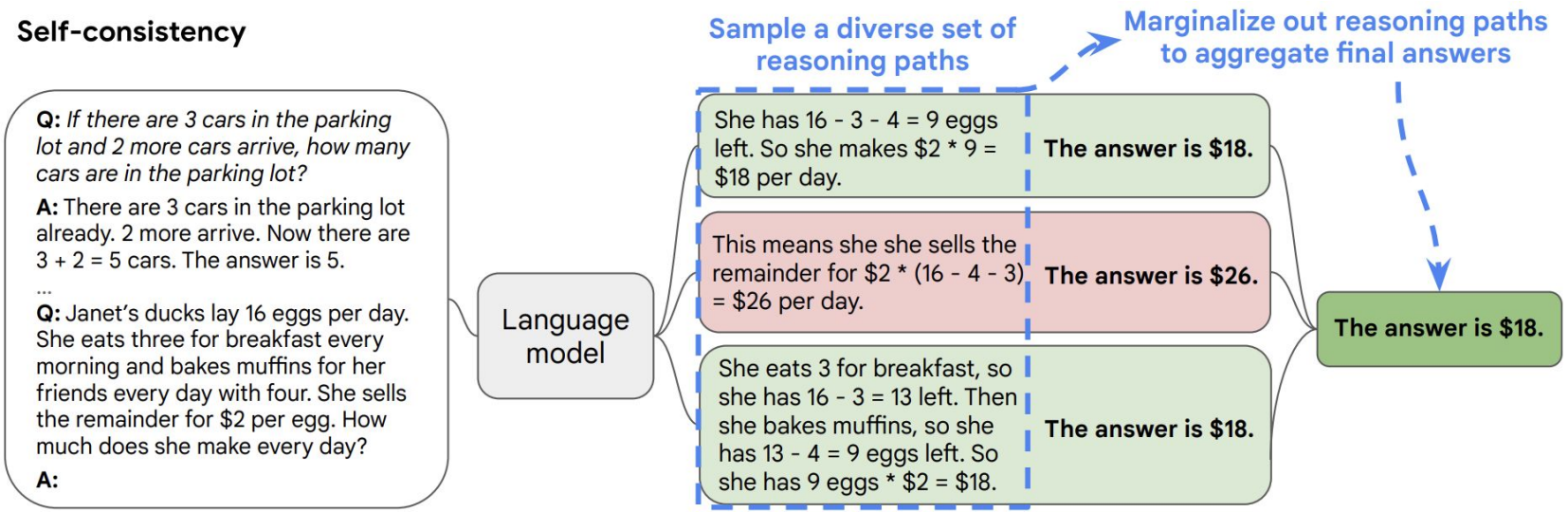
(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

- Better than zero-shot or few-shot without chain-of-thought
- But few-shot chain-of-thought is still better

Kojima et al; 2023

Self-consistency

Instead of a single reasoning explanation, sample several and marginalize out to get the final answer



$$P(a \mid q) = \sum_d P(a \mid d) P(d \mid q)$$

Wang et al; 2023

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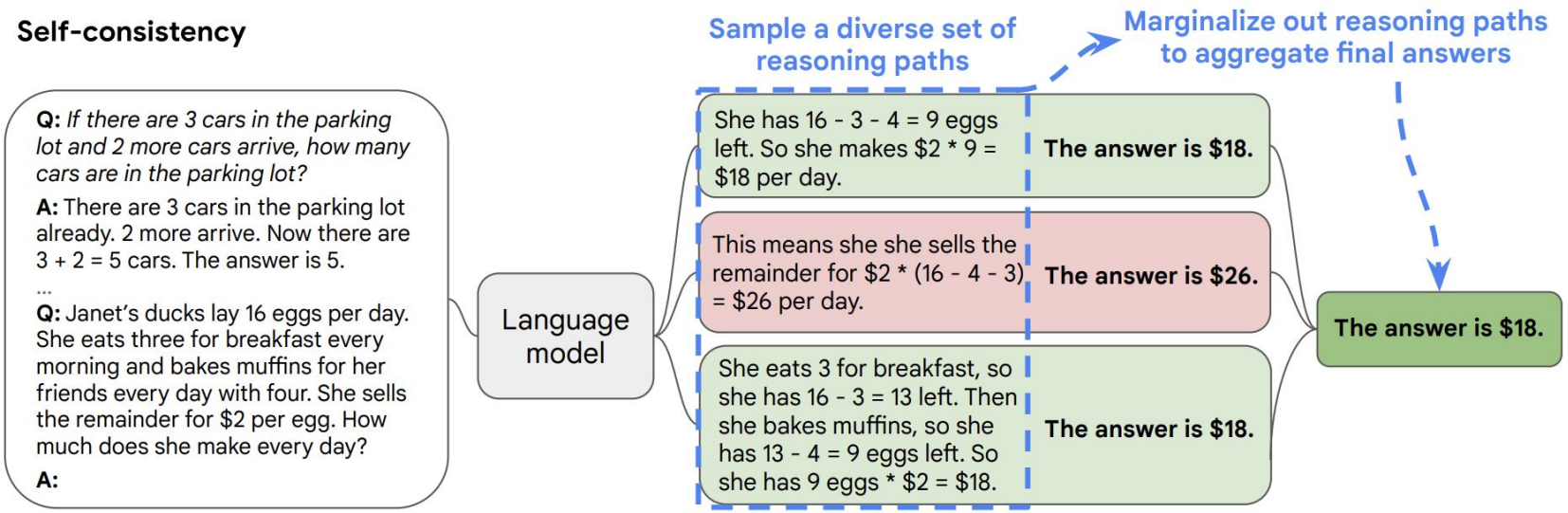


$$P(a \mid q) = \sum_d P(a \mid d) P(d \mid q)$$

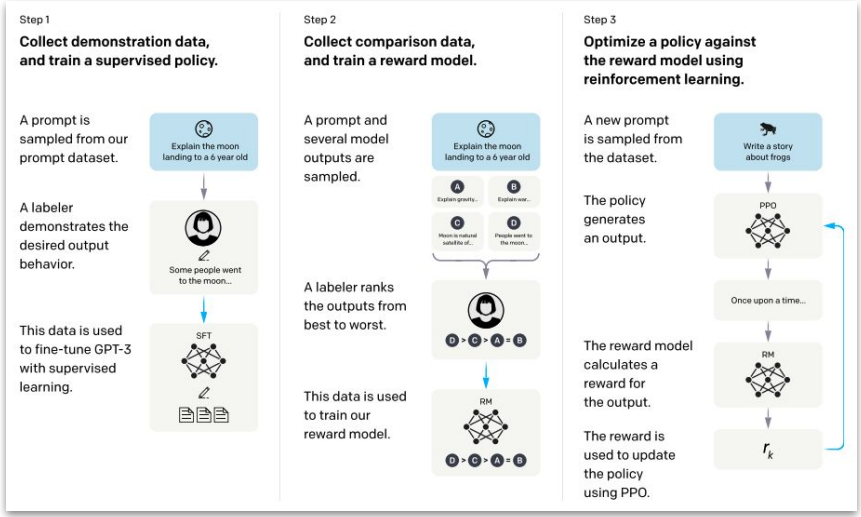
Improves GSM8K accuracy from 56.5 (chain-of-thought) to 74.4

Wang et al; 2023

Next: Training models to improve reasoning

We can do this at two different training stages:

1. Supervised finetuning (SFT)
2. Reinforcement learning (RL)



SFT for reasoning

Where do we get the training data?

1. Ask humans to write long reasoning traces

Answer the following yes/no question by reasoning step-by-step.

Can you write a whole Haiku in a single tweet?

→

A haiku is a japanese three-line poem. That is short enough to fit in 280 characters. The answer is yes.

SFT for reasoning

Where do we get the training data?

1. Ask humans to write long reasoning traces
2. Or, if you know the correct answer, generate the “correct” reasoning from a pretrained LM

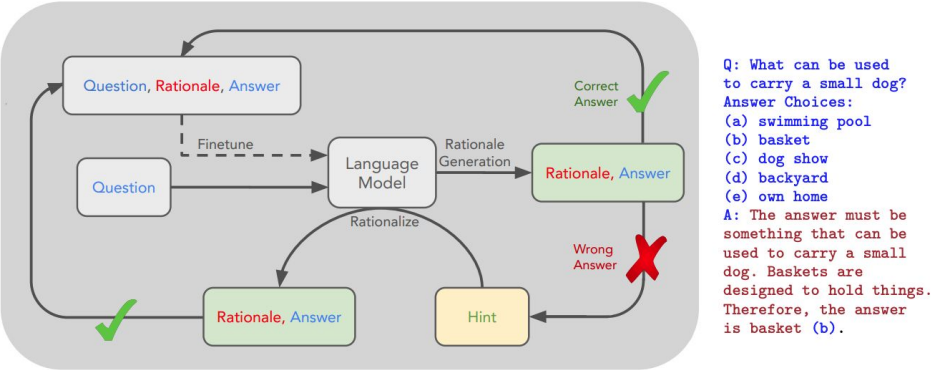


Figure 1: An overview of STaR and a STaR-generated rationale on CommonsenseQA. We indicate the fine-tuning outer loop with a dashed line. The questions and ground truth answers are expected to be present in the dataset, while the rationales are generated using STaR.

STaR: Self-Taught Reasoner, Zelikman et al, Neurips 2022

RL for reasoning

Basic idea is the same as RLHF:

- Generate samples from the LM
- Compute reward for each generated sample
- Maximize the reward by taking a gradient step

RLHF Pipeline Step 3: Reinforcement learning

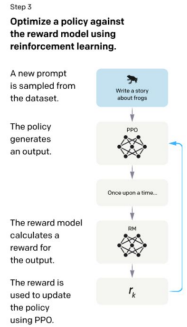
- We can estimate the expectation with samples

$$\nabla_\theta \mathbb{E}_{s\sim p_\theta}[r_\phi(s)] = \mathbb{E}_{s\sim p_\theta}[r_\phi(s)\nabla_\theta \log p_\theta(s)] \approx \frac{1}{m}\sum_{i=1}^{m} r_\phi(s_i)\nabla_\theta \log p_\theta(s_i)$$

- Update for the parameters θ:

$$\theta_{t+1} \leftarrow \theta_t + \eta \nabla_\theta \mathbb{E}_{s\sim p_\theta}[r_\phi(s)]$$

- If the reward is positive the update maximizes the (log)probability of the output (and vice versa)



InstructGPT, Ouyang et al, 2022

How to compute the reward for reasoning outputs?

For reasoning problems we often have a well-defined ground-truth answer

Solve this math problem. Show your reasoning in <think></think> tags and put the final answer in <answer></answer> tags.

A rectangle has perimeter 30 and width 5. What is its area?

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<think>
The perimeter is 30, so 2(L + 5) = 30.
Thus L + 5 = 15 and L = 10.
Area = 10 * 5 = 50.
</think>
<answer>
50
</answer>

Reward = 1

<think>
2(L + 5) = 30, so L = 15.
Area = 15 * 5 = 75.
</think>
<answer>
75
</answer>

Reward = 0

<think>
I think the answer is 50 because the rectangle seems straightforward.
</think>
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Reinforcement Learning from Verifiable Rewards (RLVR)

RLVR for other domains

1. Code generation
   ⇒ Unit tests for computing rewards

2. STEM reasoning
   ⇒ Multiple-choice questions with known answers

Deepseek-R1

Iterative application of RLVR and SFT.

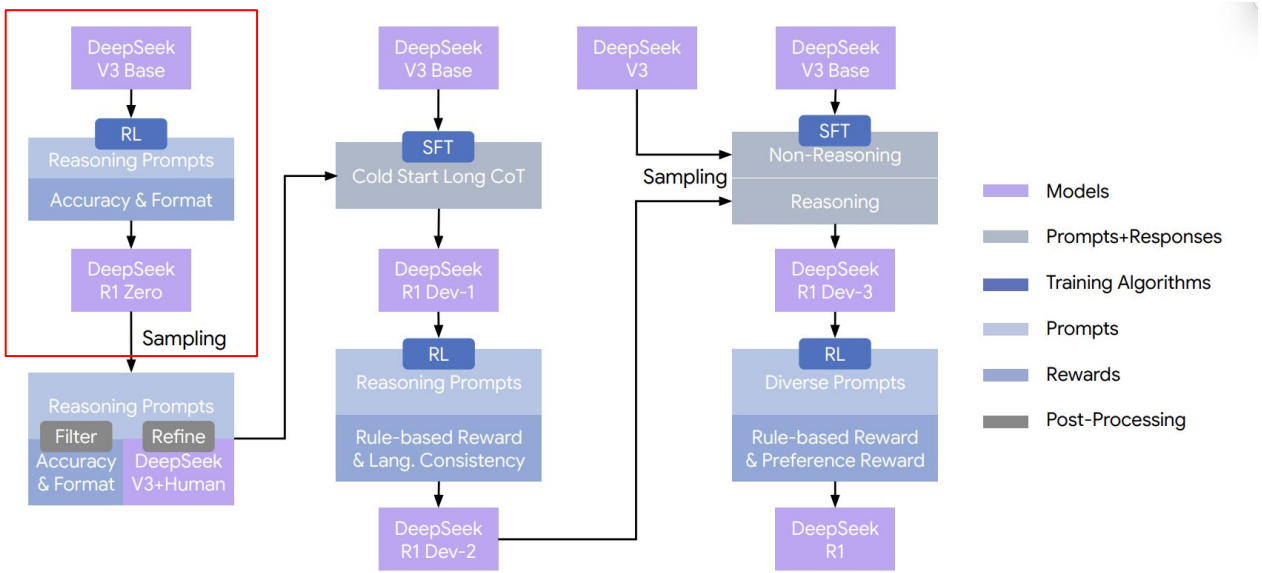


Figure 2 | The multi-stage pipeline of DeepSeek-R1. A detailed background on DeepSeek-V3 Base and DeepSeek-V3 is provided in Supplementary A.1. The models DeepSeek-R1 Dev1, Dev2, and Dev3 represent intermediate checkpoints within this pipeline.

DeepSeek-R1, 2025

Interesting phenomenon

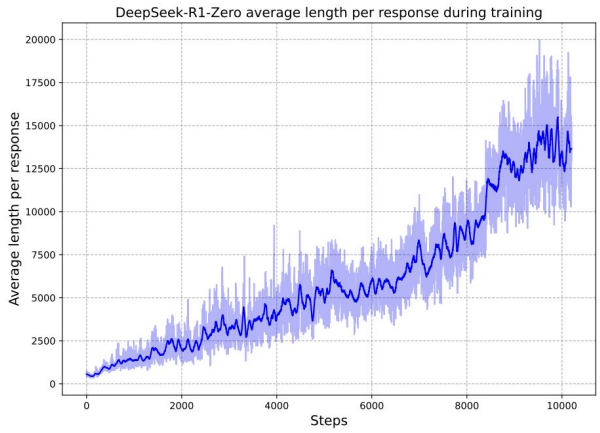
Table 1 | Template for DeepSeek-R1-Zero. prompt will be replaced with the specific reasoning question during training.

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>...</think>` and `<answer>...</answer>` tags, respectively, i.e., `<think>` reasoning process here `</think>` `<answer>` answer here `</answer>`. User: prompt. Assistant:

Table 2 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a-\sqrt{a+x}} = x$ is equal to
Response: `<think>` To solve the equation $\sqrt{a-\sqrt{a+x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a-\sqrt{a+x}}\right)^2 = x^2 \implies a - \sqrt{a+x} = x^2$. Rearrange to isolate the inner square root term: $(a-x^2)^2 = a+x \implies a^2 - 2ax^2 + (x^2)^2 = a+x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a-\sqrt{a+x}} = x$ First, let's square both sides: $a - \sqrt{a+x} = x^2 \implies \sqrt{a+x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

"aha" moment



Longer CoTs during training

Also observed: self-verification – the model checks its own work after deriving answer – and backtracking

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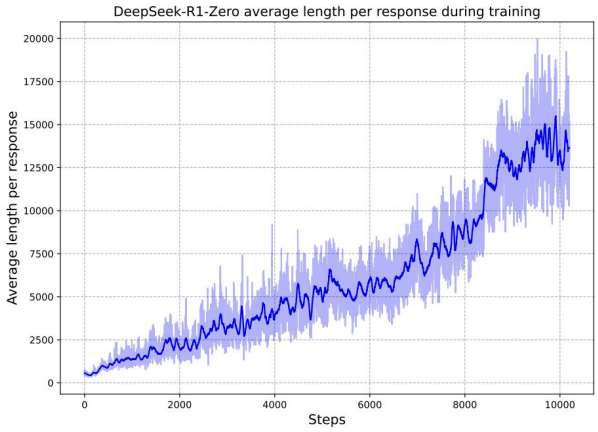

But follow up work suggests base pretrained model also has "aha" moments.

Some issues with R1-Zero

- Poor readability of reasoning traces
- Language mixing in the outputs
- Only works for verifiable tasks (math, code, mcqa)

Deepseek-R1

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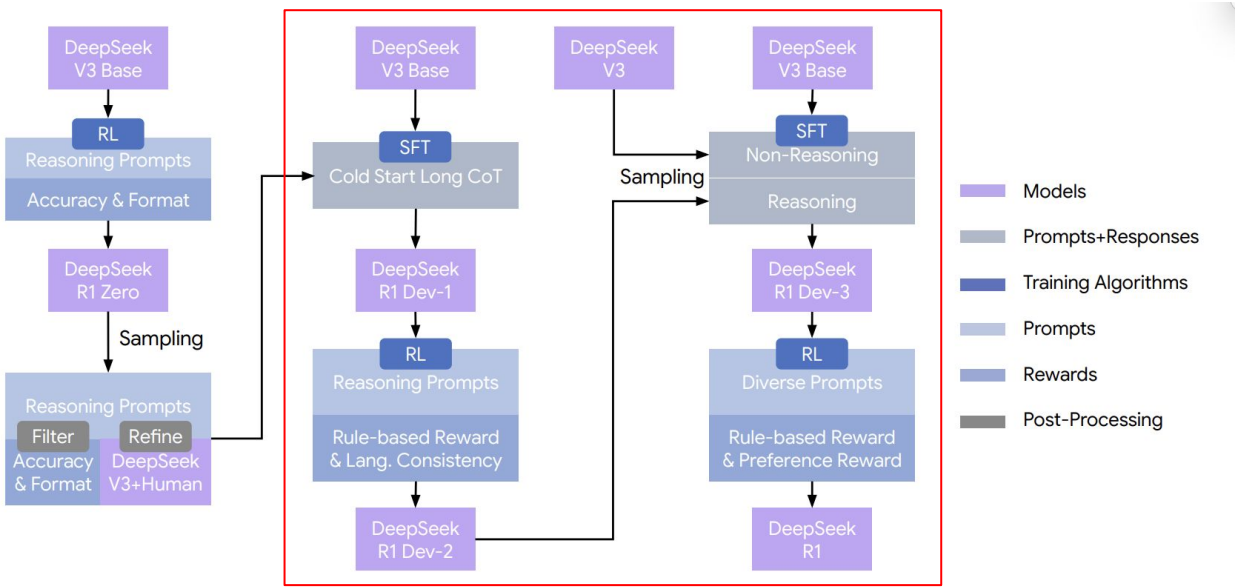

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DeepSeek-R1, 2025

RL algorithms

1. PPO
2. GRPO

RL algorithms

Recall that we are looking to maximize the expected reward:

$$\mathbb{E}_{y \sim p_\theta(\cdot|x)} \left[R(x, y) \right]$$

response
(reasoning + answer)

language model
(policy)

prompt

reward

RL algorithms

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REINFORCE: derive the gradient of this objective function to optimize via SGD

$$\nabla_\theta \mathbb{E}_{y \sim p_\theta} [R(x, y)] = \mathbb{E}_{y \sim p_\theta} [R(x, y) \nabla_\theta \log p_\theta(y|x)] \approx \frac{1}{m} \sum_{i=1}^{m} R(x, y_i) \nabla_\theta \log p_\theta(y_i|x)$$

PPO: Proximal Policy Optimization

REINFORCE can have very high gradient estimates, which leads to training instability

PPO: limit how much the policy updates in each step.

Proximal Policy Optimization

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PPO: limit how much the policy updates in each step.

$$\mathbb{E}_{y_t \sim p_\theta(\cdot|x, y_{<t})} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

Proximal Policy Optimization

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computed at
token level

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computed at token level

likelihood ratio

$$r_t(\theta) = \frac{p_\theta(y_t|x, y_{<t})}{p_{\text{old}}(y_t|x, y_{<t})}$$

p_old is the policy which generated the samples

Proximal Policy Optimization

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likelihood ratio

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advantage

$$\approx R(x, y) - V_\phi(x, y_{<t})$$

(value network)

How much better is this action than expected?

Proximal Policy Optimization

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(value network)

clipping range (typically 0.2)

Proximal Policy Optimization

REINFORCE can have very high gradient estimates, which leads to training instability

PPO: limit how much the policy updates in each step.

$$\mathbb{E}_{y_t \sim p_\theta(\cdot|x, y_{<t})} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

We can compute gradients for this in the same way as REINFORCE

Problems with PPO

- Need to train a separate value network (also an LM) in addition to the policy network

- The final reward also has a KL penalty term against the base SFT (or pretrained) LM

$$R'(x, y) = R(x, y) - \lambda D_{KL}\left(p_\theta(x, y) \| p_0(x, y)\right)$$

pretrained / SFT model

GRPO: Group Relative Policy Optimization

From an older Deepseek paper focused on math LLMs – now standard for RLVR

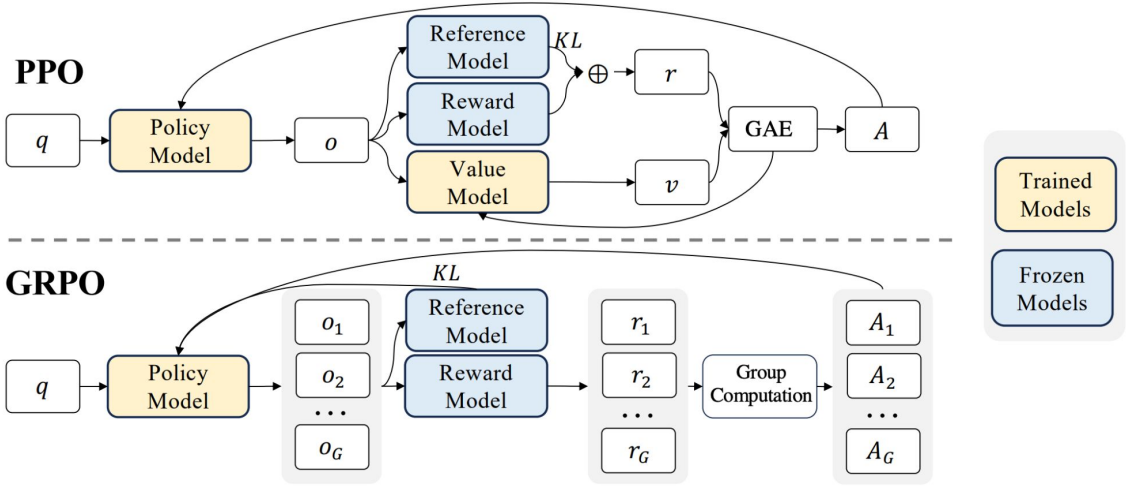

Figure 3 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the advantages from group scores.

DeepSeekMath, Shao et al, 2024

GRPO: Group Relative Policy Optimization

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Similar training objective, with key differences

DeepSeekMath, Shao et al, 2024

GRPO: Group Relative Policy Optimization

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Similar training objective, with key differences

$$\mathbb{E}_{o_1,\ldots,o_G \sim p_\theta(\cdot|x,y_{<t})} \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_{i,t} \right) - \lambda D_{KL}(p_\theta || p_0) \right]$$

group of rollouts

average across group

DeepSeekMath, Shao et al, 2024

GRPO: Group Relative Policy Optimization

From an older Deepseek paper focused on math LLMs – now standard for RLVR

Similar training objective, with key differences

$$\mathbb{E}_{o_1,\ldots,o_G \sim p_\theta(\cdot|x,y_{<t})}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta),1-\epsilon,1+\epsilon)\hat{A}_{i,t}\right) - \lambda D_{KL}(p_\theta||p_0)\right]$$

group of rollouts

average across group

advantage is computed by normalizing across group

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(R_1,\ldots,R_G)}{\text{std}(R_1,\ldots,R_G)}$$

This also means that all tokens receive exactly the same reward

DeepSeekMath, Shao et al, 2024

GRPO: Group Relative Policy Optimization

From an older Deepseek paper focused on math LLMs – now standard for RLVR

Similar training objective, with key differences

$$\mathbb{E}_{o_1,\ldots,o_G \sim p_\theta(\cdot|x,y_{<t})}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_{i,t}\right) - \lambda D_{KL}(p_\theta||p_0)\right]$$

group of rollouts

average across group

advantage is computed by normalizing across group

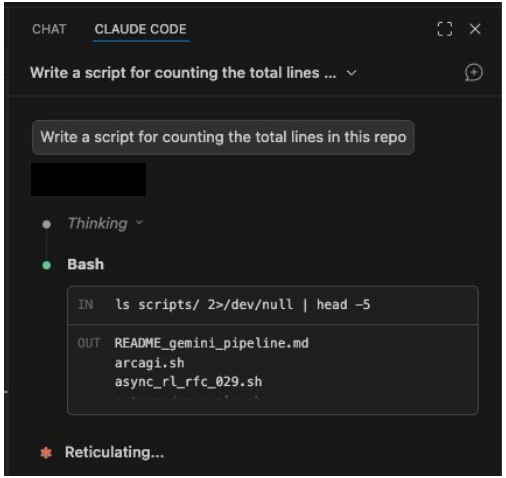
KL term is explicitly optimized in objective
(PPO folds it inside the reward)

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(R_1,\ldots,R_G)}{\text{std}(R_1,\ldots,R_G)}$$

DeepSeekMath, Shao et al, 2024

An important note

- Each post-training stage involves many datasets / types of tasks
  - Code, math, instruction following, question answering, multilingual, …
- Each of these subsets can have very different number of examples
  - E.g., 1M math prompts but only 10K coding prompts
- Important to tune the relative weights of each data component
  - A good baseline is to start with a *uniform mixture* (i.e., equal examples from each dataset)
  - But some problems are harder, so may need to sample more from such datasets

Summary

- Reasoning enables some of the most impressive applications of LLMs today
- Lots of dependencies between different post-training stages
  (but also successful recipes such as Qwen, Kimi, R1)
- Open question how *faithful* chain-of-thought is